Backpaper Examination

Fundamentals of Computing and Programming

BMath First Year

Indian Statistical Institute, Bangalore Centre

Date: 2nd January, 2026 | Time: 3 hours

Attempt all questions | Total marks: 100

Problem 1 (10 marks)

What is the output of the following program? Explain your answer in a few lines.

```c
#include<stdio.h>

void f(int n)
{
    n += 10;
}
void g(int *n)
{
    *n += 10;
}
int main(void)
{
    int x = 15;
    int y = 16;
    f(x);
    g(&y);
    printf("%d %d\n", x, y);
}
```

Question 2 (10 marks)

What is the output of the following program? Explain your answer in a few lines.

```c
#include<stdio.h>

void f(int *a, int n)
{
    int b, c, *p;

    if (*a > *(a + 1))
    {
        b = *a;
        c = *(a + 1);
    }
    else
    {
        b = *(a + 1);
        c = *a;
    }
    for (p = a + 2; p < a + n; p++)
    {
        if (*p > b)
        {
            c = b;
            b = p*;
        }
        else if (*p > c)
            c = *p;
    }
    printf("%d", c);
}

int main(void)
{
    int b[] = {23, 47, 62, 79, 99, 18, 52, 4, 86, 101};
    int *a = &b[0];
    f(a, 10);
    return 0;
}
```

Question 3 (10 marks)

What is the output of the following program?

```c
#include<stdio.h>

void move(int n, int x, int y, int z)
{
    if (n == 1)
    {
        printf("%d -> %d\n", x, y);
    }
    else
    {
        move(n-1, x, y, z);
        printf("%d -> %d", x, y);
        move(n-1, z, y, x);
    }
}
int main(void)
{
    move(3, 1, 2, 3);
    return 0;
}
```

Question 4 (10 marks)

Explain the following program in a few lines. What does it do?

```c
#include<stdio.h>

int main(int argc, char *argv[])
{
    int i;

    for(i = argc - 1; i > 0; i--)
        printf("%s ", argv[i]);
    printf("\n");
    return 0;
}
```

Question 5 (10 marks)

What is the output of the following program? Explain your answer in a few lines.

```c
#include<stdio.h>
int f(int n)
{
    int i, result = 0;
    for(i = 1; i <= n; i++)
    {
        result += 1;
        return result;
    }
}
int main(void)
{
    printf("%d\n", f(10));
    return 0;
}
```

Question 6 (10 marks)

What is the output of the following program? Explain your answer in a few lines.

```c
#include<stdio.h>
int f(unsigned x)
{
    int b;
    for(b = 0; x != 0; x >>= 1)
        if (x & 01)
            b++;
    return b;
}
int main(void)
{
    printf("%d\n", f(29));
    return 0;
}
```

Question 7 (10 marks)

What is the output of this program? Explain your answer in a few lines.

```c
#include<stdio.h>
#include<string.h>

#define F(c) ('a' <= (c) && (c) <= 'z' ? (c) - 'a' + 'A' : (c))

int main(void)
{
    int i = 0;
    char *s;
    strcpy(s, "xyzuvw");
    putchar(F(s[i++]));
    return 0;
}
```

Question 8 (10 marks)

Write a function

```c
int *largest(int a[], int n);
```

which when passed an array of length n will return a pointer to the array's largest element.

Question 9 (10 marks)

Write a function named `duplicate` which uses dynamic storage allocation to create a copy of a string. For example, the call

```c
p = duplicate(str);
```

would allocate space for a string of the same length as the string `str`, copy the contents of the string `str` into the new string using the `strcpy` function and return a pointer to it. Make duplicate return a null pointer if the memory allocation fails.

Question 10 (10 marks)

The integer 145 has the property that the sum of the factorial of its digits is equal to itself. That is, $1! + 4! + 5! = 145$. Write a program to find all such positive integers.